B.Sc. Engg. CSE 4th Semester 15 September 2025

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

FINAL SEMESTER EXAMINATION **SUMMER SEMESTER, 2023-2024**
DURATION: 3 HOURS **FULL MARKS: 200**

CSE 4405: Data and Telecommunications

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) A pseudocode for some access protocol is presented in Code Snippet 1. 2 + 5 + 5 + 3 (CO3) (PO1)

```
1  int transmitData() {
2      int k = 0;
3      while (k < MAX_RETRIES) {
4          if (isMediumFree() == 1) {  // Medium free
5              printf("Transmitting frame...\n");
6              if (isConflictDetected() == 1) {  // Conflict detected
7                  abortCurrentTransmission();
8                  sendJammingSignal();
9                  int backoff = rand() % (1 << k);
10                 wait(backoff*SLOT_TIME);
11                 k++;
12             } else {
13                 printf("Transmission successful.\n");
14                 return 1;
15             }
16         } else {
17             printf("Medium busy. Waiting...\n");
18         }
19     }
20     printf("Transmission failed after max retries.\n");
21     return -1;
22 }
```

Code Snippet 1: A pseudocode for Question 1.a

i. Which protocol is indicated by Code Snippet 1? Justify.
ii. Explain the process denoted by the code block on lines 6-11 with its necessity.
iii. What is the vulnerable time for this protocol? Explain with an appropriate figure.
iv. Is there any restriction on the frame size of the given protocol? If so, what is the restriction and why?

b) What are the different controlled access protocols? Illustrate the process of each protocol with appropriate figures. 10 (CO3) (PO1)

c) Provide the chip sequences for 9 stations using CDMA, generated from a Walsh table. 5 (CO3) (PO1)

2. a) Explain the Selective Repeat ARQ protocol. Using a suitable diagram, illustrate how the sender and receiver handle frame transmission, acknowledgments, and retransmissions in case of lost or erroneous frames. 15 (CO3) (PO1)

b) Describe the two transfer modes in High-level Data Link Control (HDLC). What are the different types of framing in HDLC? Using an appropriate figure, illustrate the HDLC frame structure and discuss the purpose of each field. 5 + 10 (CO3) (PO1)

3. a) Table 1 presents the virtual circuit (VC) forwarding tables for the switches in the sample network shown in Figure 1. Answer the following questions. 3 × 2 + 6 (CO3) (PO1)

   i. Trace the complete path (with intermediate steps) for the following data.

      1. Data with VCI13 received at port 3 in Switch S3.
      2. Data with VCI13 received at port 4 in Switch S2.

   ii. If D4 terminates its VC with D2, explain the teardown process step-by-step with the corresponding updates in the VC tables of the affected switches.

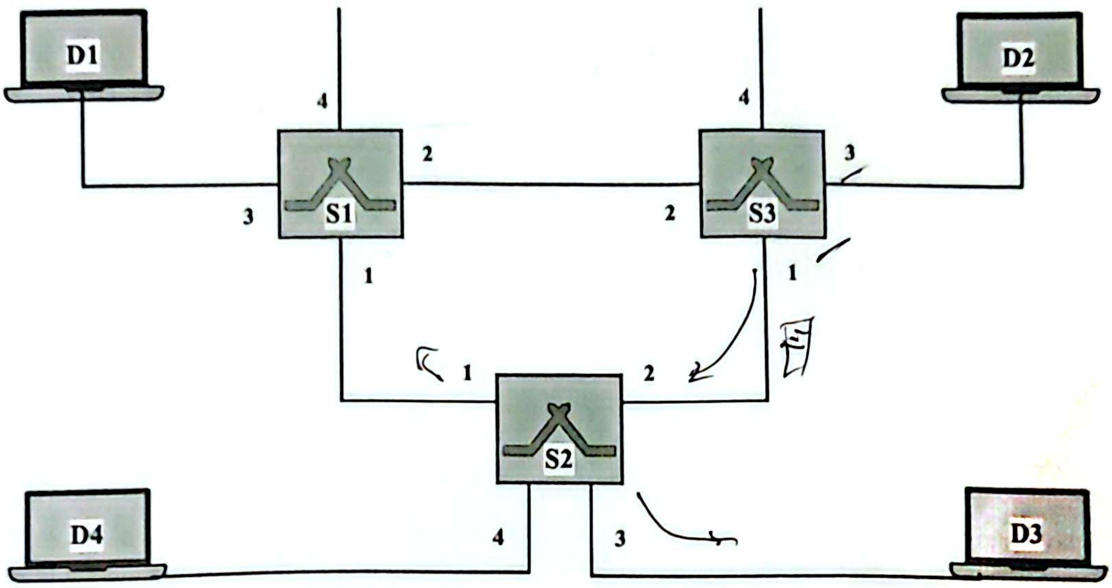

Figure 1: A sample network for Question 3.a

Table 1: Table Entries for Virtual Circuit for Question 3.a

Switch S1				Switch S2				Switch S3			
Incoming		Outgoing		Incoming		Outgoing		Incoming		Outgoing	
Port	VCI	Port	VCI	Port	VCI	Port	VCI	Port	VCI	Port	VCI
3	10	2	11	3	10	1	17	2	11	3	10
2	16	3	11	2	11	3	12	3	12	2	16
3	12	1	13	4	11	1	18	3	13	1	11
1	17	3	13	3	12	2	16	3	14	1	12
3	14	1	15	1	13	3	10	1	19	3	15
1	18	3	15	4	13	2	19	1	16	3	13
				2	12	4	13				
				4	14	3	19				
				1	15	4	11				
				3	18	4	14				

b) Describe the functionality of the Address Resolution Protocol (ARP). Explain the ARP process using an example where four devices (D1, D2, D3, and D4) are connected in a star topology through a switch, S. Now, D1 wants to communicate with D3 when it only knows D3's IP address. Use appropriate figures to support your explanation. 13 (CO3) (PO1)

c) Explain different types of addresses used in link layer protocol with examples. 5 (CO3) (PO1)

4. a) Assume the protocol X has a data packet of size 2 Bytes, of which the last 4 bits are reserved for the error handling mechanism. The protocol uses CRC for error detection. The input data before adding the CRC is $(CAB)_{16}$. If the shared divisor is $(15)_{16}$, what will be the final packet? 10 (CO3) (PO1)

b) Find the minimum Hamming distance for the detection of 6 errors and correction of 2 errors. Illustrate how burst error correction can be performed with the Hamming code. 4 + 6 (CO3) (PO1)

c) Traditionally, the Internet has used a 16-bit checksum. What are the steps at Sender side and Receiver side to calculate this traditional checksum? 10 (CO3) (PO1)

5. a) Draw the digital signals for $(7C21)_{16}$ using Unipolar NRZ, Polar NRZ-L, Polar RZ, Differential Manchester, and Pseudoternary line coding schemes on graph paper. 5×4 (CO3) (PO1)

b) What are the different categories of transmission media? Give a short description of each category. 10 (CO3) (PO1)

6. a) In GSM communication, reliable transmission is a key challenge due to the nature of the wireless channel. The major transmission problems that need to be addressed include shadowing, Rayleigh fading, time dispersion, and time alignment. Illustrate these problems with appropriate figures and discuss their possible solutions. 4×4 (CO4) (PO1)

b) In GSM, each 20 ms speech segment is coded into 260 bits during the speech coding stage (divided into three categories of 50, 132, and 78 bits based on descending importance) and is then converted into 456 bits during channel coding. Speech coding alone does not address transmission problems. Describe the channel coding and interleaving stages in detail, and explain how these stages overcome transmission problems. 7×2 (CO4) (PO1)

c) Assume that a mobile operator has been allocated a total bandwidth of 20 MHz, with each carrier having a bandwidth of 200 kHz. The operator plans to install hexagonal microcell of 500 meter radius. The operator want to cover an area of 32 Square Kilometer. Each subscriber generates 2 calls per hour with an average call duration of 3 minutes in that area. Now answer the following. 10 (CO4) (PO1)

[Note: Area of Hexagon is $\frac{3 \times \sqrt{3}}{2} R^2$. Approximation to some level is allowed while using Erlang B chart.]

i. What is the maximum number of users per cell with 2% GOS?
ii. Assume that the total number of users in that area is 10,000 and uniformly distributed in the cells. How many channels are required to give 2% GOS?

d) Interference is the major limiting factor in the performance of cellular radio systems. What are the major system generated interference and how can they be addressed? 10 (CO4) (PO1)

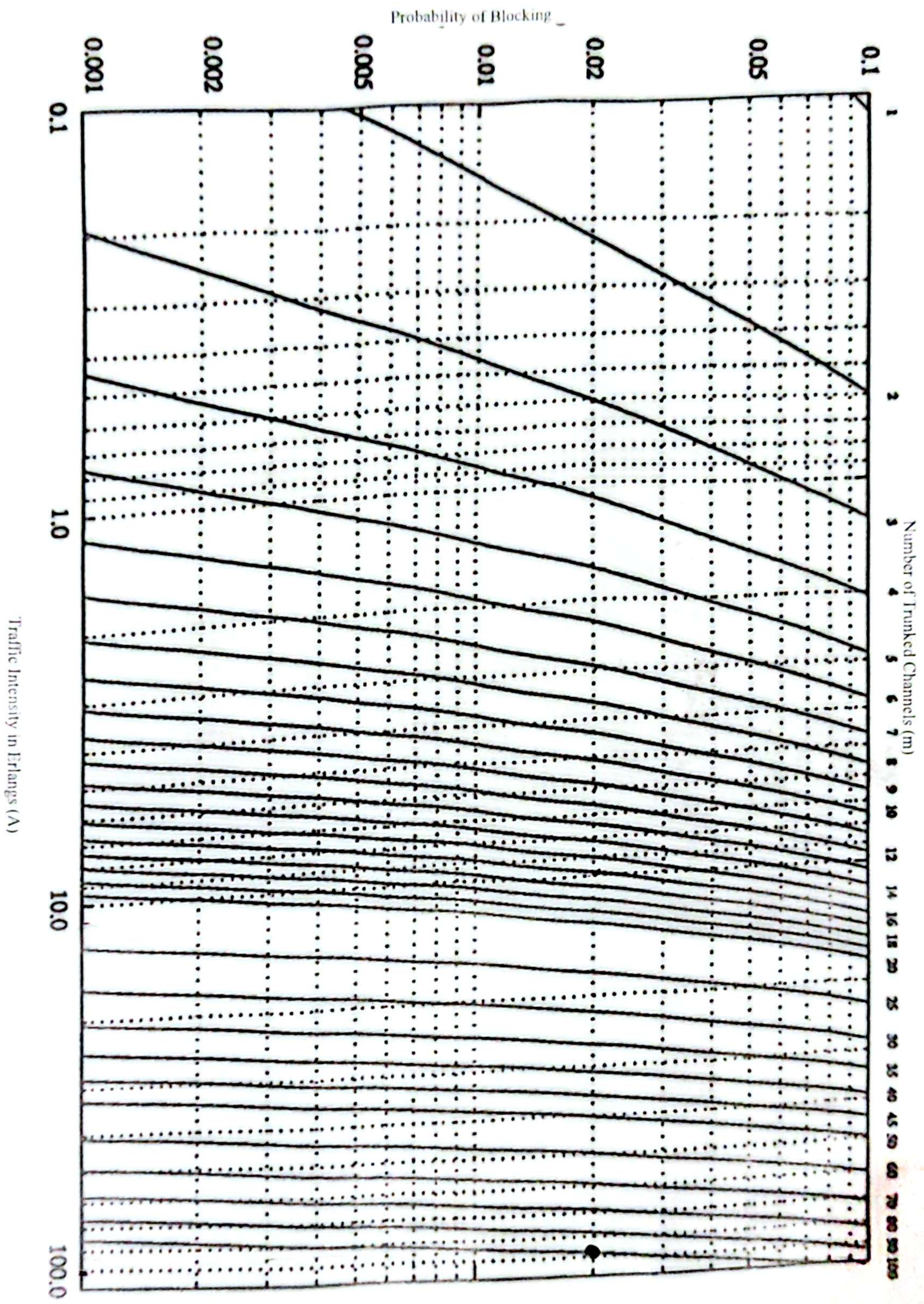

Figure 2: Erlang B Chart